

### **# import lib**

```
import pandas as pd
from nltk import word_tokenize
from nltk.corpus import stopwords
from string import punctuation
from nltk.stem import PorterStemmer
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import classification_report
```

### **# load the data**

```
data = pd.read_csv("spam.csv")
print(data)
```

### **# check and handle null data**

```
print(data.isnull().sum())
```

### **# data cleaning**

```
sw = stopwords.words("english")
ps = PorterStemmer()
def clean_text(txt):
    txt = txt.lower()
    txt = word_tokenize(txt)
    txt = [t for t in txt if t not in punctuation]
    txt = [t for t in txt if t not in sw]
    txt = [ps.stem(t) for t in txt]
    txt = " ".join(txt)
    return txt
```

```
data["clean_text"] = data["text"].apply(clean_text)
print(data)
```

### **# features and target**

```
cv = CountVectorizer()
vector = cv.fit_transform(data["clean_text"])
features = pd.DataFrame(vector.toarray(),
columns=cv.get_feature_names_out())
print(features)
target = data["label"]
print(target)
```

### **# train and test**

```
x_train, x_test, y_train, y_test = train_test_split(features.values, target)
```

### **# model**

```
model = MultinomialNB()
model.fit(x_train, y_train)
```

### **# classification\_report**

```
y_pred = model.predict(x_test)
cr = classification_report(y_test, y_pred)
print(cr)
```

### **# prediction**

```
text =
c_text =
v_text =
ans =
print(ans))
```